

2.6 ETL / TSE FACTOR & ECHO TRAIN LENGTH

ACQUIRING MORE LINES PER EXCITATION

ETL (Echo Train Length) or TSE Factor determines how many k-space lines are collected after a single excitation. Longer echo trains speed up the scan but can affect image quality. The technologist must balance speed and image fidelity.



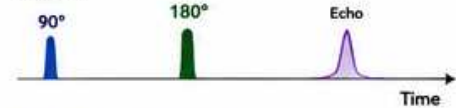
KEY CONCEPT

Longer ETL = faster scan (fewer excitations)
But more T2 decay, blurring and artifacts.
Choose the highest ETL that maintains
diagnostic image quality.

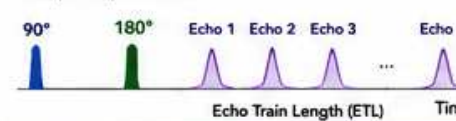
↑ Increase / Improve
↓ Decrease / Reduce
— No / Minimal Effect

1 WHAT IS ETL / TSE FACTOR?

In Spin Echo (SE):
One excitation → one
echo → one line of
k-space.



In Turbo Spin Echo (TSE):
One excitation → many echoes →
multiple k-space lines.



Definitions

ETL / TSE Factor
= Number of echoes
(k-space lines) collected
per excitation.

Higher ETL
= Fewer excitations
= Shorter scan time.

But...
= More T2 decay during
echo train.

2 EFFECT OF ETL ON IMAGE QUALITY

ETL	Relative Scan Time	T2 Decay / Blurring	SNR	Artifacts (Blurring, Ghosting, Crosstalk)	Overall Image Quality
Short (2-4)	↓	Minimal	Good	Minimal	Excellent
Moderate (8-16)	Moderate	Mild	Good	Mild	Good
Long (>16)	Short	Significant	Lower	More likely	Fair to Poor

Rule of Thumb: Use the highest ETL that still provides diagnostic quality.

3 EFFECT OF ETL ON SCAN TIME

$$\text{Scan Time} \propto \frac{\# \text{ of Phase Encodes}}{\text{ETL}} \times \text{Other Factors}$$

Therefore:

↑ Increase ETL → Decrease scan time

↓ Decrease ETL → Increase scan time

Example (Same Parameters)

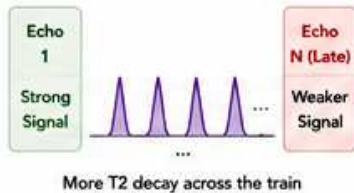
ETL	Relative Scan Time	Change
2	8x	—
4	4x	↓ 50%
8	2x	↓ 75%
16	1x	↓ 87.5%
32	0.5x	↓ 93.8%



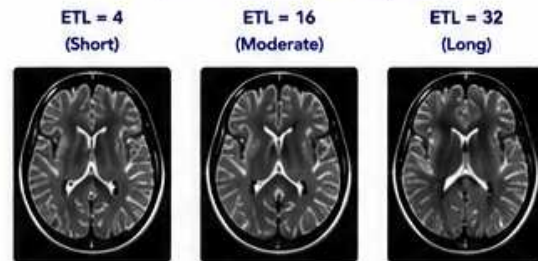
Doubling ETL ≈ halves scan time (approximate).

4 HOW ECHO TRAIN (ETL) AFFECTS IMAGE APPEARANCE

As the echo train progresses, more
T2 decay occurs → signal loss.
Later echoes (high ETL) have less
signal → more blurring.



T2-Weighted Axial Brain (ETL Comparison)



Less Blurring (Better Detail) → More Blurring (Less Detail)

5 ETL & ARTIFACTS

Higher ETL can increase:

- Blurring
- Ghosting
- Echo crosstalk
- B0 / B1 inhomogeneity artifacts
- SAR (specific absorption rate)

T2-Weighted Spine (Sagittal)



6 ETL RECOMMENDATIONS BY APPLICATION

Application / Region	Typical ETL Range	Comments
Brain (Routine T2)	16 – 32	Higher ETL for faster scans, watch for blurring
Spine	12 – 24	Moderate ETL, balance speed & detail
MSK (Joints)	8 – 16	Lower ETL for fine detail
Abdomen	16 – 24	Use with fat sat as appropriate
Cardiac (T2 TSE)	8 – 16	Lower ETL to reduce artifacts
High Resolution	4 – 8	Detail prioritized over speed

Always evaluate final image quality — not just the numbers.

7 ETL VS. OTHER PARAMETERS

Parameter	Increase ETL	Effect on Image	Effect on Scan Time	Other Notes
NEX (Averages)	↑ SNR	Improves SNR	↑ Longer	Useful with high ETL
Bandwidth	↑ Less Noise	May slightly improve	—	Higher BW still best
Matrix	—	No direct effect	—	Resolution independent
Slice Thickness	—	No direct effect	—	—
Field Strength d	—	May increase SNR	—	3T more sensitive to artifacts

ETL works with other parameters — balance the whole prescription.

8 PRACTICAL GUIDELINES

- Know the clinical goal: Speed (screening) vs. detail (high resolution).
- Start moderate, then adjust: Increase ETL until quality starts to drop.
- Look for early signs of degradation: Blurred edges, loss of CSF detail, ghosting.
- Balance patient factors: Longer ETL reduces time and motion risk.

9 COMMON ETL SETTINGS (GENERAL GUIDE)

Application	Typical ETL Range	Notes
Brain T2 TSE	16 – 32	Often 16 or 24
Spine T2 TSE	12 – 24	Adjust for detail
Knee T2 TSE	8 – 16	Lower for fine structures
Shoulder T2 TSE	8 – 16	Lower to reduce artifacts
Abdomen T2 TSE	16 – 24	Use with fat sat
Pelvis T2 TSE	12 – 24	Balance speed & clarity

10 ENGINEER CHECKLIST

- ☒ Is ETL appropriate for the clinical goal?
- ☒ Is image quality acceptable (detail, sharpness)?
- ☒ Are artifacts within acceptable limits?
- ☒ Is scan time reasonable for the patient?
- ☒ Is SNR adequate?
- ☒ Have other parameters been optimized?



KEY TAKEAWAY

ETL (TSE Factor) is a powerful tool to reduce scan time. Use the highest ETL that delivers diagnostic image quality. Balance speed, detail, artifacts, and patient comfort.